

Bayesian inference for non-Gaussian simultaneous autoregressive models with missing data

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ABSTRACT

Standard simultaneous autoregressive (SAR) models typically assume normally distributed errors, an assumption often violated in real-world datasets that frequently exhibit non-normal, skewed, or heavy-tailed characteristics. New SAR models are proposed to capture these non-Gaussian features. The spatial error model (SEM), a widely used SAR-type model, is considered. Three novel SEMs are introduced, extending the standard Gaussian SEM. These extensions incorporate Student's t -distributed errors to accommodate heavy-tailed behaviour, one-to-one transformations of the response variable to address skewness, or a combination of both. Variational Bayes (VB) estimation methods are developed for these models, and the framework is further extended to handle missing response data under the missing not at random (MNAR) mechanism. Standard VB methods perform well with complete datasets; however, handling missing data requires a hybrid VB (HVB) approach, which integrates a Markov chain Monte Carlo (MCMC) sampler to generate missing values. The proposed VB methods are evaluated using both simulated and real-world datasets, demonstrating their robustness and effectiveness in dealing with non-Gaussian data and missing data in spatial models. Although the method is demonstrated using SAR models, the proposed model specifications and estimation approaches are widely applicable to various types of models for handling non-Gaussian data with missing values.

1. Introduction

Simultaneous autoregressive (SAR) models are a broad class of spatial statistical models commonly used for analysing spatially correlated lattice data, with applications spanning ecology, social sciences, and finance (Cressie, 1993; Chi and Zhu, 2008; Calabrese et al., 2017; Ver Hoef et al., 2018). Examples of SAR models include spatial error model (SEM), spatial autoregressive models (SAM), and spatial Durbin model (SDM), among others (Anselin, 1988; Cressie, 1993). Similar to other conventional spatial models, SAR models typically assume Gaussian-distributed data; see Besag (1974), Anselin (1988), Cressie (1993). However, this normality assumption is frequently violated in practice, as real-world datasets often exhibit non-normal characteristics such as skewness and heavy-tailed distributions.

Several studies have examined the estimation of SAR models when the underlying error distributions are non-Gaussian, leading to methods that do not rely on strong distributional assumptions. For example, quasi-maximum likelihood estimators (QMLE) have been proposed for the SAM (Lee, 2004), while various Generalised Method of Moments (GMM) approaches have been developed for different types of SAR models, which are robust to unknown heteroskedasticity and deviations from Gaussianity (Kelejian and

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Prucha, 1998; Lee and Liu, 2010; Breitung and Wigger, 2018). However, these approaches require the response variable to be fully observed and therefore cannot accommodate missing data.

Missing values frequently arise in spatial datasets. When estimating SAR models, ignoring missing response values can lead to inconsistency and bias (Wang and Lee, 2013; Benedetti et al., 2020). Various estimation methods for SAR models with randomly missing data have been well developed (see LeSage and Pace (2004), Wang and Lee (2013), Suesse and Zammit-Mangion (2017), Suesse (2018), Wijayawardhana et al. (2024)). However, research on estimating SAR models when data are missing not at random (MNAR) remains relatively scarce. Flores-Lagunes and Schnier (2012) proposed a GMM estimator for this setting, while Doğan and Taşpinar (2018) and Seya et al. (2021) employed the Metropolis–Hastings (MH) algorithm.

Bayesian estimation of SAR models using Markov chain Monte Carlo (MCMC) methods can be computationally demanding, particularly when the data contain missing values (Wijayawardhana et al., 2025). Variational Bayes (VB) provides a computationally efficient alternative, yet it remains underutilised in the context of SAR models. For instance, Wu (2018) and Bansal et al. (2021) applied VB to fully observed SAR models, while Wijayawardhana et al. (2025) proposed hybrid VB (HVB) approaches for SAR models with missing data. Moreover, Bayesian inference methods, including both VB and MCMC, offer full posterior uncertainty quantification, in contrast to GMM, which yields only point estimates with asymptotic standard errors.

Our article makes several key contributions aimed at addressing limitations in the current literature on SAR models, particularly in handling non-Gaussian responses, missing data, and the computational challenges arising from these complex scenarios. First, we introduce novel SAR models designed to capture non-Gaussian characteristics of the response variable, such as skewness and heavy tails. We extend conventional Gaussian SAR models in two ways: by incorporating Student's t -distributed errors to address heavy-tailedness, and by applying the Yeo–Johnson (YJ) transformation (Yeo and Johnson, 2000) to the response variable to handle skewness (see Section 2.2). Unlike the traditional two-step approach, where skewness is first reduced using a transformation such as YJ or Box–Cox and then a model is fitted, our integrated framework combines these steps within the SAR structure. These innovations yield three novel SAR models that jointly capture skewness and heavy-tailedness, making them well-suited for complex real-world data. While we demonstrate these models using SEM, yielding three SEM variants, the framework is broadly applicable to other SAR models.

Second, we propose efficient VB methods for estimating these new SEMs, both with and without missing values in the response variable. For SEMs with missing values, we use an extended version of the HVB algorithm of Wijayawardhana et al. (2025) to handle non-Gaussian response variables and missing data under the MNAR mechanism. While the proposed HVB algorithm is an approximation compared to the MCMC methods (Doğan and Taşpinar, 2018; Seya et al., 2021) and Hamiltonian Monte Carlo (HMC) (Neal, 2011), it offers substantial computational advantages. Our simulation study in Section 6 shows that the HVB algorithm is more than ten times faster than the HMC implementation in RStan (Stan Development Team, 2023) for a dataset of $n = 625$ observations with approximately 50% missing responses. Furthermore, Sections 7 and 8 show that the proposed HVB approach scales efficiently to much larger datasets, for which HMC becomes computationally impractical.

The rest of the paper is organised as follows. Section 2 introduces the new SEMs, and Section 3 discusses SEMs with missing responses. In Section 4, we present the VB methods for estimating SEMs with missing data. Section 5 discusses Bayesian model comparison. Sections 6 and 7 present two simulation studies, and Section 8 applies the SEMs to a real-world dataset. Section 9 summarises the main findings and outlines directions for future work. The paper also has an online supplement with additional technical details. The R code to reproduce all simulations and real-world analyses is available at <https://github.com/AnjanaWijayawardhana/Non-Gaussian-SAR-with-Missing-Data-VB>.

2. Simultaneous autoregressive models

In this section, we use our proposed framework to extend conventional Gaussian SAR models to non-Gaussian settings. We showcase the extension using spatial error models (SEMs), a widely adopted form of SAR-type models.

Let $\mathbf{y}^* = (y_1^*, y_2^*, \dots, y_n^*)^\top$ be the $n \times 1$ vector of response variable observed at n spatial locations $s_1, \dots, s_n$, $\mathbf{X}$ be the $n \times (r+1)$ design matrix containing the covariates, and $\mathbf{W}$ be the $n \times n$ spatial weight matrix. The SEM is given by

$$\mathbf{y}^* = \mathbf{X}\boldsymbol{\beta} + (\mathbf{I}_n - \rho\mathbf{W})^{-1}\mathbf{e}, \quad (1)$$

where $\mathbf{e}$ is the $n \times 1$ vector of error terms, $\boldsymbol{\beta} = (\beta_0, \beta_1, \dots, \beta_r)^\top$ is the $(r+1) \times 1$ vector of fixed effects parameters, ρ is the spatial autocorrelation parameter, which measures the strength and direction of spatial dependence (LeSage and Pace, 2009), and $\mathbf{I}_n$ is the $n \times n$ identity matrix. Section 2.1 discusses SEMs with Gaussian and Student's t errors. Section 2.2 discusses two SEMs with the Yeo–Johnson transformation (Yeo and Johnson, 2000).

2.1. SEMs with Gaussian and student's t errors

The conventional SEM assumes Gaussian error terms $\mathbf{e} \sim N(\mathbf{0}, \sigma_e^2 \mathbf{I}_n)$ in the model presented in Eq. (1), where σ_e^2 is the error variance parameter. This assumption implies that the distribution of $\mathbf{y}^*$ is also Gaussian, with the mean vector $\mathbf{X}\boldsymbol{\beta}$ and covariance matrix $\sigma_e^2(\mathbf{A}^\top \mathbf{A})^{-1}$, where $\mathbf{A} = \mathbf{I}_n - \rho\mathbf{W}$. We call this model the SEM-Gau.

We now discuss SEM- t , which assumes that the error terms $\mathbf{e}$ in Eq. (1) follow a Student's t -distribution. Student's t -distribution has been widely used to model heavy-tailed data in popular statistical models, such as regression (Lange et al., 1989; Fernández and Steel, 1999) and mixed-effects models (Pinheiro et al., 2001; Wang and Lin, 2014). Huang et al. (2021) applied Student's t errors to spatial autoregressive scalar-on-function regression models.

Table 1

The definitions of $\mathbf{r}$, $\mathbf{M}$, and the parameters ϕ for SEM-Gau, SEM-t, YJ-SEM-Gau, and YJ-SEM-t. For SEM-Gau and SEM-t, the density of the response vector conditional on ξ follows a multivariate Gaussian distribution with mean $\mathbf{X}\beta$ and covariance matrix $\sigma_e^2 \mathbf{M}^{-1}$, where $\mathbf{A} = \mathbf{I}_n - \rho \mathbf{W}$.

Model	ϕ	$\mathbf{r}$	$\mathbf{M}$
SEM-Gau	$(\beta^\top, \sigma_e^2, \rho)^\top$	$\mathbf{y} - \mathbf{X}\beta$	$\mathbf{A}^\top \mathbf{A}$
SEM-t	$(\beta^\top, \sigma_e^2, \rho, \nu)^\top$	$\mathbf{y} - \mathbf{X}\beta$	$\mathbf{A}^\top \Sigma_\tau^{-1} \mathbf{A}$
YJ-SEM-Gau	$(\beta^\top, \sigma_e^2, \rho, \gamma)^\top$	$t_\gamma(\mathbf{y}) - \mathbf{X}\beta$	$\mathbf{A}^\top \mathbf{A}$
YJ-SEM-t	$(\beta^\top, \sigma_e^2, \rho, \nu, \gamma)^\top$	$t_\gamma(\mathbf{y}) - \mathbf{X}\beta$	$\mathbf{A}^\top \Sigma_\tau^{-1} \mathbf{A}$

Let $e_i \sim t_\nu(0, \sigma_e^2)$, where e_i is the i th element of the error vector $\mathbf{e}$ in Eq. (1) and $t_\nu(0, \sigma_e^2)$ is the Student's t distribution with ν degrees of freedom ($\nu \geq 3$), mean zero, and scale parameter σ_e^2 . These Student's t errors, e_i for $i = 1, \dots, n$, can be expressed as a scale mixture of normals (Chan et al., 2019):

$$\begin{aligned} e_i | \tau_i, \sigma_e^2 &\sim N(0, \sigma_e^2 \tau_i), \\ \tau_i | \nu &\sim IG\left(\frac{\nu}{2}, \frac{\nu}{2}\right), \quad i = 1, \dots, n \end{aligned} \quad (2)$$

where $IG(a, b)$ denotes the inverse gamma distribution with shape parameter a and rate parameter b . The conditional distribution of errors presented in Eq. (2), can be written in vector notation as

$$\mathbf{e} | \tau, \sigma_e^2 \sim N(0, \sigma_e^2 \Sigma_\tau), \quad (3)$$

where the matrix $\Sigma_\tau = \text{diag}(\tau_1, \dots, \tau_n)$. We now incorporate the error structure from Eq. (3) into the SEM described in Eq. (1). As a result, the distribution of $\mathbf{y}^*$ conditional on the latent vector τ is also Gaussian, with the mean vector $\mathbf{X}\beta$ and covariance matrix $\sigma_e^2 (\mathbf{A}^\top \Sigma_\tau^{-1} \mathbf{A})^{-1}$, where $\mathbf{A} = \mathbf{I}_n - \rho \mathbf{W}$. This model is referred to as the SEM-t. The log-likelihood function of $\mathbf{y}^*$ for both the SEM-Gau and SEM-t is given by:

$$\log p_{\mathbf{y}^*}(\mathbf{y}^* | \xi) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma_e^2) + \frac{1}{2} \log |\mathbf{M}| - \frac{1}{2\sigma_e^2} \mathbf{r}^\top \mathbf{M} \mathbf{r}, \quad (4)$$

where for the SEM-Gau, $\xi = \phi$, while for the SEM-t, $\xi = (\phi^\top, \tau^\top)^\top$. The definitions for $\mathbf{r}$, $\mathbf{M}$, and ϕ are given in Table 1.

2.2. SEMs with transformations

This section extends the SEM-Gau and SEM-t in Eq. (1) by applying the YJ transformation element-wise to the response vector $\mathbf{y}^*$.

Let y_i be the i th element of the response vector in the SEM with YJ transformation. We assume that y_i is modelled as

$$y_i = t_\gamma^{-1}(y_i^*), \quad i = 1, \dots, n, \quad (5)$$

where y_i^* is the i th element of the response vector in the SEM given in Eq. (1), with either Gaussian or Student's t -distributed errors, and $t_\gamma^{-1}(\cdot)$ is the inverse function of the YJ transformation. For $0 < \gamma < 2$, it is defined as:

$$y_i = t_\gamma^{-1}(y_i^*) = \begin{cases} (y_i^* \gamma + 1)^{1/\gamma} - 1 & \text{if } y_i^* \geq 0 \\ 1 - (-(2 - \gamma)y_i^* + 1)^{1/(2-\gamma)} & \text{if } y_i^* < 0. \end{cases} \quad (6)$$

As the YJ transformation makes the data more symmetric and less skewed, applying its inverse transformation moves the data away from symmetry. This implies that $\mathbf{y} = t_\gamma^{-1}(\mathbf{y}^*) = (t_\gamma^{-1}(y_1^*), t_\gamma^{-1}(y_2^*), \dots, t_\gamma^{-1}(y_n^*))^\top$ becomes an asymmetric, non-Gaussian response variable. In vector notation, the SEM with the YJ transformation (YJ-SEM) is defined as:

$$\mathbf{y} = t_\gamma^{-1}(\mathbf{y}^*) = t_\gamma^{-1}(\mathbf{X}\beta + (\mathbf{I}_n - \rho \mathbf{W})^{-1} \mathbf{e}). \quad (7)$$

Using the standard technique for transforming random variables, it is straightforward to show that the density of $\mathbf{y}$ for the YJ-SEM is:

$$p_{\mathbf{y}}(\mathbf{y} | \xi) = p_{\mathbf{y}^*}(t_\gamma(\mathbf{y})) \prod_{i=1}^n \left(\frac{dt_\gamma(y_i)}{dy_i} \right), \quad i = 1, \dots, n, \quad (8)$$

where $p_{\mathbf{y}^*}(\cdot)$ is the density of $\mathbf{y}^*$, conditional on the parameter vector ξ ; see Section 2.1, and $\frac{dt_\gamma(y_i)}{dy_i}$ is the derivative of the YJ transformation with respect to y_i ; see Equation (S2) in Section S1 of the online supplement. The vector ξ denotes the parameters of the YJ-SEM defined in Eq. (7).

The YJ-SEM with Gaussian errors (YJ-SEM-Gau) assumes that the error term in Eq. (7) follows a normal distribution: $\mathbf{e} \sim N(0, \sigma_e^2 \mathbf{I}_n)$, where σ_e^2 is a variance parameter. The YJ-SEM with Student's t -distributed errors (YJ-SEM-t) assumes that the error

term follows the structure given in Eq. (3). The log-likelihood function of $\mathbf{y}$ for YJ-SEM-Gau and YJ-SEM-t is expressed as:

$$\begin{aligned} \log p_{\mathbf{y}}(\mathbf{y} | \xi) = & -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma_e^2) + \frac{1}{2} \log |\mathbf{M}| - \frac{1}{2\sigma_e^2} \mathbf{r}^\top \mathbf{M} \mathbf{r} \\ & + \sum_{i=1}^n \log \left(\frac{dt_{\gamma}(y_i)}{dy_i} \right), \end{aligned} \quad (9)$$

where for the YJ-SEM-Gau, $\xi = \phi$, while for the YJ-SEM-t, $\xi = (\phi^\top, \tau^\top)^\top$. The definitions for $\mathbf{r}$, $\mathbf{M}$, and ϕ are given in Table 1. Note that $y_i = y_i^*$ for $i = 1, \dots, n$, corresponds to an identity transformation in both SEM-Gau and SEM-t.

The three SEMs, SEM-Gau, SEM-t, and YJ-SEM-Gau, are special cases of the YJ-SEM-t. The SEM-Gau is a YJ-SEM-t where the degrees of freedom parameter (ν) equals ∞ and the YJ parameter (γ) equals 1, while the YJ-SEM-Gau is a YJ-SEM-t with $\nu = \infty$ and $\gamma \neq 1$. Finally, the SEM-t is also a YJ-SEM-t with $\gamma = 1$.

3. SEMs with missing responses

This section develops the formulation of SEMs in settings where response values are missing and introduces the underlying missing data mechanisms, focusing in particular on the missing not at random (MNAR) mechanism.

Consider that the response vector $\mathbf{y}$ of an SEM presented in Sections 2.1 and 2.2 contains missing values. Let $\mathbf{y}_o$ denote the subset of $\mathbf{y}$ with n_o observed units, and $\mathbf{y}_u$ denote the subset with n_u unobserved (missing) units. The complete response vector can then be written as $\mathbf{y} = (\mathbf{y}_o^\top, \mathbf{y}_u^\top)^\top$.

We define a binary missingness indicator vector $\mathbf{m} = (m_1, \dots, m_n)^\top$, where

$$m_i = \begin{cases} 1, & \text{if } y_i \text{ is missing,} \\ 0, & \text{if } y_i \text{ is observed.} \end{cases}$$

In the presence of missing responses, the vector $\mathbf{r}$ and the matrices $\mathbf{X}$ and $\mathbf{M}$, as defined in Table 1, are partitioned into corresponding observed and unobserved components as follows:

$$\mathbf{r} = \begin{pmatrix} \mathbf{r}_o \\ \mathbf{r}_u \end{pmatrix}, \quad \mathbf{X} = \begin{pmatrix} \mathbf{X}_o \\ \mathbf{X}_u \end{pmatrix}, \quad \mathbf{M} = \begin{pmatrix} \mathbf{M}_{oo} & \mathbf{M}_{ou} \\ \mathbf{M}_{uo} & \mathbf{M}_{uu} \end{pmatrix}, \quad (10)$$

where $\mathbf{r}_o$ and $\mathbf{r}_u$ are subvectors of $\mathbf{r}$ corresponding to the observed and unobserved responses, respectively. Similarly, $\mathbf{X}_o$ and $\mathbf{X}_u$ represent the design matrices associated with the observed and unobserved responses, while $\mathbf{M}_{oo}$, $\mathbf{M}_{ou}$, $\mathbf{M}_{uo}$, and $\mathbf{M}_{uu}$ denote the corresponding submatrices of $\mathbf{M}$.

3.1. Missing data mechanism and MNAR assumption

The missing data mechanism is characterised by the joint probability mass function of the missingness indicator vector $\mathbf{m} = (m_1, \dots, m_n)^\top$, where each element m_i denotes whether the corresponding response y_i is missing ($m_i = 1$) or observed ($m_i = 0$). Let $p(m_i | y_i, \mathbf{x}_i^*, \boldsymbol{\psi})$ denote the probability that y_i is missing, possibly depending on itself and on a set of covariates $\mathbf{x}_i^* = (1, x_{i1}, \dots, x_{iq})^\top$. The parameter vector $\boldsymbol{\psi} = (\boldsymbol{\psi}_x^\top, \boldsymbol{\psi}_y^\top)^\top$ consists of the fixed effect parameter vector associated with covariates $\mathbf{x}_i^*$, denoted as $\boldsymbol{\psi}_x = (\psi_0, \psi_1, \psi_2, \dots, \psi_q)^\top$, and the fixed effect parameter corresponding to y , denoted as ψ_y .

Three main types of missing-data mechanisms are commonly distinguished following Rubin (1976): (i) *missing completely at random* (MCAR), where the probability of missingness is independent of both the observed and unobserved data and on observed covariates, that is, $p(m_i | y_i, \mathbf{x}_i^*, \boldsymbol{\psi}) = p(m_i | \boldsymbol{\psi})$; (ii) *missing at random* (MAR), where the probability of missingness may depend on observed covariates but not on the observed and unobserved responses, so that $p(m_i | y_i, \mathbf{x}_i^*, \boldsymbol{\psi}) = p(m_i | \mathbf{x}_i^*, \boldsymbol{\psi})$; and (iii) *missing not at random* (MNAR), where the probability of missingness depends on the response variable itself and observed covariates. Under MNAR, the term $p(m_i | y_i, \mathbf{x}_i^*, \boldsymbol{\psi})$ cannot be simplified further.

The conditional probability mass function of m_i under MNAR, $p(m_i | y_i, \mathbf{x}_i^*, \boldsymbol{\psi})$ can be modelled using any binary-response model, such as a logistic or probit regression. In this paper, we model the probability that the i th response is missing via a logistic regression:

$$p(m_i = 1 | y_i, \mathbf{x}_i^*, \boldsymbol{\psi}) = \frac{\exp(\mathbf{x}_i^{*\top} \boldsymbol{\psi}_x + y_i \psi_y)}{1 + \exp(\mathbf{x}_i^{*\top} \boldsymbol{\psi}_x + y_i \psi_y)}, \quad (11)$$

so that the joint probability mass function of vector $\mathbf{m}$ is

$$p(\mathbf{m} | \mathbf{y}, \mathbf{X}^*, \boldsymbol{\psi}) = \prod_{i=1}^n \frac{\exp\{(\mathbf{x}_i^{*\top} \boldsymbol{\psi}_x + y_i \psi_y) m_i\}}{1 + \exp(\mathbf{x}_i^{*\top} \boldsymbol{\psi}_x + y_i \psi_y)}, \quad (12)$$

where $\mathbf{X}^* = (\mathbf{x}_1^{*\top}, \dots, \mathbf{x}_n^{*\top})^\top$ denotes the design matrix for the missingness model, with dimension $n \times (q + 1)$.

Table 2

The term $h(\xi, \psi, \mathbf{y}_u) = p(\mathbf{y} | \xi)p(\mathbf{m} | \mathbf{y}, \psi)p(\xi)p(\psi)$, and the total number of model parameters s (including the length of the latent vector τ if exists) for the SEM-Gau, SEM-t, YJ-SEM-Gau, and YJ-SEM-t with missing responses. The density $p(\tau | \phi) = \prod_{i=1}^n p(\tau_i | v)$ represents the n independent inverse gamma latent variables used in SEMs with Student- t errors (see Eq. (2)).

Model	$h(\xi, \psi, \mathbf{y}_u)$	s
SEM-Gau	$p(\mathbf{y} \phi)p(\mathbf{m} \mathbf{y}, \psi)p(\phi)p(\psi)$	$r + q + 5$
SEM-t	$p(\mathbf{y} \tau, \phi)p(\mathbf{m} \mathbf{y}, \psi)p(\tau \phi)p(\phi)p(\psi)$	$r + q + 6 + n$
YJ-SEM-Gau	$p(\mathbf{y} \phi)p(\mathbf{m} \mathbf{y}, \psi)p(\phi)p(\psi)$	$r + q + 6$
YJ-SEM-t	$p(\mathbf{y} \tau, \phi)p(\mathbf{m} \mathbf{y}, \psi)p(\tau \phi)p(\phi)p(\psi)$	$r + q + 7 + n$

3.2. Joint modelling of $\mathbf{y}$ and $\mathbf{m}$

In our study, the primary objective is the estimation of the parameters of the proposed SEMs, denoted by ξ , which govern the distribution of the response vector $\mathbf{y}$. However, under the MNAR mechanism, the missingness process is not ignorable, and valid inference requires modelling the joint distribution of $(\mathbf{y}, \mathbf{m})$ or, equivalently, the joint estimation of ξ and ψ (Wijayawardhana et al., 2025). The joint distribution of $\mathbf{y}$ and $\mathbf{m}$ is denoted by $p(\mathbf{y}, \mathbf{m} | \xi, \psi)$. Using the selection model factorisation (Little and Rubin, 2019), $p(\mathbf{y}, \mathbf{m} | \xi, \psi)$ can be factorised as

$$p(\mathbf{y}, \mathbf{m} | \xi, \psi) = p(\mathbf{m} | \mathbf{y}, \psi)p(\mathbf{y} | \xi), \quad (13)$$

where $p(\mathbf{y} | \xi)$ represents the likelihood of the SEM, and $p(\mathbf{m} | \mathbf{y}, \psi)$ denotes the joint probability mass function of missingness indicators defined in Eq. (12). The log-likelihood of $\mathbf{y}$ for the different SEMs, along with their full model specifications, is provided in Section 2.

It is important to note that the factorisation in Eq. (13) is a standard and widely applicable decomposition for jointly modelling a missingness mechanism and a statistical model with missing data, and it applies to any SAR-type model; see Section S4 of the online supplement. This factorisation enables practical inference for SAR models with missing values under MNAR. In Section 4, we discuss the estimation of the joint posterior distribution of the model parameters (ξ, ψ) and $\mathbf{y}_u$ using the proposed variational Bayes method.

4. Variational Bayes inference for SEMs with missing data

Section 4.1 provides an overview of the variational Bayes (VB) inference framework for estimating SEMs with missing responses. Section 4.2 then introduces an efficient hybrid variational Bayes (HVB) algorithm. For completeness, Section S2 of the online supplement presents the VB algorithm for SEMs with fully observed data.

4.1. Variational Bayes inference under missing responses

Variational Bayes (VB) inference approximates a posterior distribution by formulating it as an optimisation problem, offering a more computationally efficient alternative to the computationally intensive Markov chain Monte Carlo (MCMC) methods for complex statistical models. The standard VB approach for SEMs with missing responses is described below.

We assume that some values in the response vector of an SEM are missing under the MNAR mechanism, as described in Section 3. Under this assumption, Bayesian inference requires estimating the joint posterior distribution of the model parameters ξ and ψ , based on the joint likelihood defined in Eq. (13). Since the subvector $\mathbf{y}_u$ of $\mathbf{y} = (\mathbf{y}_o^\top, \mathbf{y}_u^\top)^\top$ is unobserved, it must also be inferred. Consequently, Bayesian inference involves estimating the joint posterior distribution of ξ , ψ , and $\mathbf{y}_u$. For the SEM-Gau and YJ-SEM-Gau, $\xi = \phi$, while for the SEM-t and YJ-SEM-t, $\xi = (\phi^\top, \tau^\top)^\top$; see Table 1 for the definition of ϕ for each SEM.

Let the joint posterior distribution of ξ , ψ , and $\mathbf{y}_u$ be denoted as $p(\xi, \psi, \mathbf{y}_u | \mathbf{y}_o, \mathbf{m})$. We denote the prior distributions of ξ and ψ by $p(\xi)$ and $p(\psi)$, respectively. Using the selection model factorisation in Eq. (13), the joint posterior distribution $p(\xi, \psi, \mathbf{y}_u | \mathbf{y}_o, \mathbf{m})$ is given by

$$p(\xi, \psi, \mathbf{y}_u | \mathbf{y}_o, \mathbf{m}) \propto p(\mathbf{m} | \mathbf{y}, \psi)p(\mathbf{y} | \xi)p(\xi)p(\psi). \quad (14)$$

We define $h(\xi, \psi, \mathbf{y}_u) = p(\mathbf{y} | \xi)p(\mathbf{m} | \mathbf{y}, \psi)p(\xi)p(\psi)$. The term $h(\xi, \psi, \mathbf{y}_u)$ for each SEM with missing responses, along with the total number of parameters to be estimated (s) for each model, are provided in Table 2.

We consider the variational approximation $q_\lambda(\xi, \psi, \mathbf{y}_u)$, indexed by the variational parameter λ to approximate the joint posterior $p(\xi, \psi, \mathbf{y}_u | \mathbf{y}_o, \mathbf{m})$ in Eq. (14). The VB approach approximates this posterior distribution by minimising the Kullback–Leibler (KL) divergence between $q_\lambda(\xi, \psi, \mathbf{y}_u)$ and $p(\xi, \psi, \mathbf{y}_u | \mathbf{y}_o, \mathbf{m})$, which is defined as

$$\begin{aligned} \text{KL}(\lambda) &= \text{KL}(q_\lambda(\xi, \psi, \mathbf{y}_u) || p(\xi, \psi, \mathbf{y}_u | \mathbf{y}_o, \mathbf{m})) \\ &= \int \log \left(\frac{q_\lambda(\xi, \psi, \mathbf{y}_u)}{p(\xi, \psi, \mathbf{y}_u | \mathbf{y}_o, \mathbf{m})} \right) q_\lambda(\xi, \psi, \mathbf{y}_u) d\xi d\psi d\mathbf{y}_u. \end{aligned} \quad (15)$$

Minimising the KL divergence in Eq. (15) is equivalent to maximising the evidence lower bound (ELBO) on the logarithm of the marginal likelihood, $\log p(\mathbf{y}_o, \mathbf{m})$, denoted by $\mathcal{L}(\lambda)$, where $p(\mathbf{y}_o, \mathbf{m}) = \int p(\mathbf{y}_o, \mathbf{m} \mid \xi, \psi, \mathbf{y}_u) p(\mathbf{y}_u \mid \xi, \psi) p(\xi, \psi) d\xi d\psi d\mathbf{y}_u$ (Blei et al., 2017). The ELBO can be written as an expectation with respect to the variational distribution and is given by

$$\mathcal{L}(\lambda) = \mathbb{E}_{q_\lambda} [\log h(\xi, \psi, \mathbf{y}_u) - \log q_\lambda(\xi, \psi, \mathbf{y}_u)], \quad (16)$$

where $\mathbb{E}_{q_\lambda}[\cdot]$ denotes the expectation with respect to q_λ . Table 2 provides expressions for $h(\xi, \psi, \mathbf{y}_u)$ for different SEMs.

To maximise ELBO given in Eq. (16) with respect to variational parameters, λ , the stochastic gradient ascent (SGA) method is used (Nott et al., 2012; Titsias and Lázaro-Gredilla, 2014, 2015). The SGA method updates the initial value for λ (say $\lambda^{(0)}$) according to the iterative scheme,

$$\lambda^{(t+1)} = \lambda^{(t)} + \mathbf{a}^{(t)} \circ \widehat{\nabla_\lambda \mathcal{L}(\lambda^{(t)})}, \quad (17)$$

where $\widehat{\nabla_\lambda \mathcal{L}(\lambda)}$ is an unbiased estimate of the true gradient $\nabla_\lambda \mathcal{L}(\lambda)$ of the ELBO in Eq. (16), $\mathbf{a}^{(t)}$ ($t = 0, 1, \dots$) is a vector of adaptive step sizes, obtained using the ADADELTA method (Zeiler, 2012), described in Section S3 of the online supplement to facilitate rapid convergence of the SGA algorithm. The symbol $\circ$ represents the element-wise product of two vectors. The updating of Eq. (17) is done until a stopping criterion is satisfied. Further details on the SGA method are provided in Section S2 of the online supplement.

In standard VB inference, the missing data vector, $\mathbf{y}_u$, is treated as a set of parameters, and the joint posterior distribution of ξ , ψ , and $\mathbf{y}_u$ is approximated using the joint variational distribution $q_\lambda(\xi, \psi, \mathbf{y}_u)$. However, Wijayawardhana et al. (2025) demonstrated that this joint approximation is often insufficiently flexible to capture the posterior distributions of model parameters and $\mathbf{y}_u$ simultaneously, due to the complex dependencies among the spatially correlated elements of $\mathbf{y}_u$. To address this limitation, Section 4.2 introduces an alternative hybrid algorithm that effectively combines VB optimisation with MCMC sampling.

4.2. Hybrid variational Bayes

This section extends the hybrid VB (HVB) algorithm of Wijayawardhana et al. (2025) to efficiently handle missing values $\mathbf{y}_u$ in non-Gaussian SEMs proposed in Section 2. The approach employs a Gaussian variational approximation with a factor covariance structure (Ong et al., 2018) to approximate the posterior distribution of the parameters, while employing an MCMC sampler to generate samples of $\mathbf{y}_u$ from its conditional distribution, given $\mathbf{y}_o$, $\mathbf{m}$, ξ and ψ . We first factorise the joint variational distribution that approximates ξ , ψ , and $\mathbf{y}_u$, $q_\lambda(\xi, \psi, \mathbf{y}_u)$ as follows:

$$q_\lambda(\xi, \psi, \mathbf{y}_u) = p(\mathbf{y}_u \mid \mathbf{y}_o, \mathbf{m}, \xi, \psi) q_\lambda^0(\xi, \psi), \quad (18)$$

where $p(\mathbf{y}_u \mid \mathbf{y}_o, \mathbf{m}, \xi, \psi)$ represents the conditional distribution of $\mathbf{y}_u$, given the observed data ($\mathbf{y}_o$, $\mathbf{m}$), and the model parameters (ξ , ψ). The term $q_\lambda^0(\xi, \psi)$ is the variational approximation used to approximate the posterior distribution of ξ and ψ . The ELBO in Eq. (16) can be rewritten by substituting the factorised variational distribution $q_\lambda(\xi, \psi, \mathbf{y}_u)$ from Eq. (18). By applying Bayes' rule and simplifying, we obtain

$$\mathcal{L}(\lambda) = \mathbb{E}_{q_\lambda} [\log p(\mathbf{y}_o, \mathbf{m} \mid \xi, \psi) + \log p(\xi, \psi) - \log q_\lambda^0(\xi, \psi)] = \mathcal{L}^0(\lambda), \quad (19)$$

where $\mathcal{L}^0(\lambda)$ is the ELBO resulting from approximating only the posterior distribution of parameters (ξ , ψ), that is, $p(\xi, \psi \mid \mathbf{y}_o, \mathbf{m})$, directly using the variational distribution $q_\lambda^0(\xi, \psi)$; see Wijayawardhana et al. (2025) for the complete proof.

We assume that the variational distribution, $q_\lambda^0(\xi, \psi)$, follows a multivariate Gaussian distribution with a factor covariance structure (Ong et al., 2018), $q_\lambda^0(\xi, \psi) \sim N((\xi^\top, \psi^\top)^\top; \mu, \mathbf{B}\mathbf{B}^\top + \mathbf{D}^2)$, where μ is an $s \times 1$ vector of variational means, $\mathbf{B}$ is an $s \times p$ matrix with the upper triangular elements set to zero ($p \ll s$), and $\mathbf{D}$ is an $s \times s$ diagonal matrix with positive diagonal elements $\mathbf{d} = (d_1, \dots, d_s)^\top$. For all SEMs with missing values, the value of s is provided in Table 2. The vector of variational parameters is given by $\lambda = (\mu^\top, \text{vech}(\mathbf{B})^\top, \mathbf{d}^\top)^\top$, where the 'vech' operator vectorises a matrix by stacking its columns from left to right while removing all the elements above the diagonal (the super-diagonal elements) of the matrix.

We now describe how the SGA method is used in the proposed HVB algorithm. At each iteration, an unbiased estimate of the ELBO gradient, $\nabla_\lambda \mathcal{L}(\lambda)$, denoted $\widehat{\nabla_\lambda \mathcal{L}(\lambda)}$, needs to be computed for the SGA update given in Eq. (17). This is achieved using the reparameterisation trick (Kingma and Welling, 2013). First, samples are generated from the variational distribution $q_\lambda^0(\xi, \psi)$ by drawing $\delta^0 = (\eta^0, \epsilon^0) \sim N(\mathbf{0}, \mathbf{I}_{s+p})$, where η^0 is a p -dimensional vector and ϵ^0 is an s -dimensional vector. Next, we compute $(\xi^\top, \psi^\top)^\top = r^0(\delta^0, \lambda) = \mu + \mathbf{B}\eta^0 + \mathbf{d}\epsilon^0$. The density of δ^0 is denoted by $f_{\delta^0}(\delta^0)$. Let $\delta = (\delta^0^\top, \mathbf{y}_u^\top)^\top$, where the product density is expressed as $f_\delta(\delta) = f_{\delta^0}(\delta^0)p(\mathbf{y}_u \mid \mathbf{y}_o, \mathbf{m}, r^0(\delta^0, \lambda))$. Finally, the transformation from δ to the parameter and augmented missing value space is given by $((\xi^\top, \psi^\top)^\top, \mathbf{y}_u^\top)^\top = t(\delta, \lambda) = (r^0(\delta^0, \lambda)^\top, \mathbf{y}_u^\top)^\top = ((\mu + \mathbf{B}\eta^0 + \mathbf{d}\epsilon^0)^\top, \mathbf{y}_u^\top)^\top$.

The reparameterisation gradient of the ELBO in Eq. (19) is derived by differentiating under the integral sign as shown below:

$$\nabla_\lambda \mathcal{L}(\lambda) = \mathbb{E}_{f_\delta} \left[\frac{dt^0(\delta^0, \lambda)^\top}{d\lambda} (\nabla_{(\xi^\top, \psi^\top)^\top} \log h(\xi, \psi, \mathbf{y}_u) - \nabla_{(\xi^\top, \psi^\top)^\top} \log q_\lambda^0(\xi, \psi)) \right], \quad (20)$$

where $\frac{dt^0(\delta^0, \lambda)}{d\lambda}$ is the derivative of the transformation $r^0(\delta^0, \lambda) = \mu + \mathbf{B}\eta^0 + \mathbf{d}\epsilon^0$ with respect to the variational parameters $\lambda = (\mu^\top, \text{vech}(\mathbf{B})^\top, \mathbf{d}^\top)^\top$; see Section S5.1 of the online supplement for the proof. The expressions for $\frac{dt^0(\delta^0, \lambda)}{d\lambda}$ and $\nabla_{(\xi^\top, \psi^\top)^\top} \log q_\lambda^0(\xi, \psi)$ can be found in Section S5.2 of the online supplement. In addition, the expressions for $\nabla_{(\xi^\top, \psi^\top)^\top} \log h(\xi, \psi, \mathbf{y}_u)$ are provided in Section S5.3 of the online supplement for different SEMs.

Algorithm 1 Hybrid variational Bayes (HVB) algorithm.

-
- 1: Initialise $\lambda^{(0)} = (\mu^{(0)}, \text{vech}(\mathbf{B})^{(0)}, \mathbf{d}^{(0)})^\top$ and set $t = 0$
 - 2: **repeat**
 - 3: Generate $(\eta^{0(t)}, \epsilon^{0(t)}) \sim N(\mathbf{0}, \mathbf{I}_{s+p})$
 - 4: Generate $(\xi^{(t)}, \psi^{(t)})^\top \sim q_{\lambda^{(t)}}^0(\xi, \psi)$ using its reparameterised representation.
 - 5: Generate $\mathbf{y}_u^{(t)} \sim p(\mathbf{y}_u | \mathbf{y}_o, \mathbf{m}, \xi^{(t)}, \psi^{(t)})$
 - 6: Construct unbiased estimates $\widehat{\nabla_\mu \mathcal{L}(\lambda)}$, $\widehat{\nabla_{\text{vech}(\mathbf{B})} \mathcal{L}(\lambda)}$, and $\widehat{\nabla_{\mathbf{d}} \mathcal{L}(\lambda)}$ using Equations (S46), (S47) and (S48) in Section S5.2 of the online supplement.
 - 7: Set adaptive learning rates $\mathbf{a}_\mu^{(t)}$, $\mathbf{a}_{\text{vech}(\mathbf{B})}^{(t)}$ and $\mathbf{a}_{\mathbf{d}}^{(t)}$, using ADADELTA described in Section S3 of the online supplement.
 - 8: Set $\mu^{(t+1)} = \mu^{(t)} + \mathbf{a}_\mu^{(t)} \circ \widehat{\nabla_\mu \mathcal{L}(\lambda^{(t)})}$.
 - 9: Set $\text{vech}(\mathbf{B})^{(t+1)} = \text{vech}(\mathbf{B})^{(t)} + \mathbf{a}_{\text{vech}(\mathbf{B})}^{(t)} \circ \widehat{\nabla_{\text{vech}(\mathbf{B})} \mathcal{L}(\lambda^{(t)})}$.
 - 10: Set $\mathbf{d}^{(t+1)} = \mathbf{d}^{(t)} + \mathbf{a}_{\mathbf{d}}^{(t)} \circ \widehat{\nabla_{\mathbf{d}} \mathcal{L}(\lambda^{(t)})}$.
 - 11: Set $\lambda^{(t+1)} = (\mu^{(t+1)}, \text{vech}(\mathbf{B})^{(t+1)}, \mathbf{d}^{(t+1)})^\top$, and $t = t + 1$
 - 12: **until** some stopping rule is satisfied
-

Algorithm 2 MCMC steps within the t^{th} iteration of the HVB algorithm.

-
- 1: Initialise missing values $\mathbf{y}_{u,0} \sim p(\mathbf{y}_u | \xi^{(t)}, \mathbf{y}_o)$
 - 2: **for** $i = 1, \dots, N_1$ **do**
 - 3: Sample $\tilde{\mathbf{y}}_u$ from the proposal distribution $p(\tilde{\mathbf{y}}_u | \xi^{(t)}, \mathbf{y}_o)$.
 - 4: Sample u from uniform distribution, $u \sim \mathcal{U}(0, 1)$
 - 5: Calculate $a = \min\left(1, \frac{p(\mathbf{m} | \tilde{\mathbf{y}}_u, \psi^{(t)})}{p(\mathbf{m} | \mathbf{y}_{i-1}, \psi^{(t)})}\right)$, where $\tilde{\mathbf{y}} = (\mathbf{y}_o^\top, \tilde{\mathbf{y}}_u^\top)^\top$ and $\mathbf{y}_{i-1} = (\mathbf{y}_o^\top, \mathbf{y}_{u,i-1}^\top)^\top$
 - 6: **if** $a > u$ **then**
 - 7: $\mathbf{y}_{u,i} = \tilde{\mathbf{y}}_u$
 - 8: **else**
 - 9: $\mathbf{y}_{u,i} = \mathbf{y}_{u,i-1}$
 - 10: **end if**
 - 11: **end for**
 - 12: Output $\mathbf{y}_u^{(t)} = \mathbf{y}_{u,N_1}$
-

Algorithm 1 outlines the HVB algorithm. For all four SEMs with missing data, the conditional distribution of missing data, $p(\mathbf{y}_u | \mathbf{y}_o, \mathbf{m}, \xi, \psi)$ is not available in closed form. To sample from $p(\mathbf{y}_u | \mathbf{y}_o, \mathbf{m}, \xi, \psi)$ in step 5 of Algorithm 1, we use the MCMC steps outlined in Algorithm 2.

The MCMC steps in Algorithm 2 generate samples from the proposal distribution $p(\tilde{\mathbf{y}}_u | \xi^{(t)}, \mathbf{y}_o)$, which follows a multivariate Gaussian with the mean vector given by $\mathbf{X}_u \beta - \mathbf{M}_{uu}^{-1} \mathbf{M}_{uo} \mathbf{r}_o$ and the covariance matrix given by $\sigma_e^2 \mathbf{M}_{uu}^{-1}$, for both SEM-Gau and SEM-t; see Table 1 and Eq. (10) for details on the partitioning of $\mathbf{r}$, $\mathbf{X}$ and $\mathbf{M}$ across different SEMs.

For YJ-SEM-Gau and YJ-SEM-t, the proposals are generated in two steps. First, we sample $\tilde{\mathbf{y}}_u^*$ from the conditional distribution $p(\tilde{\mathbf{y}}_u^* | \xi^{(t)}, \mathbf{y}_o^*)$, which follows a multivariate Gaussian with the mean vector given by $\mathbf{X}_u \beta - \mathbf{M}_{uu}^{-1} \mathbf{M}_{uo} (\mathbf{y}_o^* - \mathbf{X}_o \beta)$ and the covariance matrix given by $\sigma_e^2 \mathbf{M}_{uu}^{-1}$, where $\mathbf{y}_o^* = t_\gamma(\mathbf{y}_o)$. Then, we apply the inverse Yeo-Johnson (YJ) transformation to obtain the final proposal: $\tilde{\mathbf{y}}_u = t_\gamma^{-1}(\tilde{\mathbf{y}}_u^*)$.

As n and n_u increase, the HVB algorithm implemented using the MCMC scheme outlined in Algorithm 2 fails to estimate the parameters accurately due to a low acceptance rate. To address this, we partition $\mathbf{y}_u$ into k blocks and update one block at a time. The MCMC steps for sampling the missing values one block at a time are detailed in Algorithm S2 in Section S5.4 of the online supplement. We refer to the modified HVB algorithm with block-wise updates as HVB-AllB (which uses MCMC steps in Algorithm S2), and the original version without block-wise updates as HVB-NoB (which uses MCMC steps in Algorithm 2) in the following sections.

5. Bayesian model comparison

To assess the model fit of different SEMs in the subsequent sections of this study, we employ extended versions of the standard Deviance Information Criterion (DIC; Spiegelhalter et al. (2002)) as proposed by Celeux et al. (2006). This section first outlines DIC methods for models without missing data, followed by those for models with missing data.

Table 3
Prior distributions of model parameters.

Parameter	β	ω'	ρ'	ν'	γ'	ψ
Prior distribution	$N(\mathbf{0}, \sigma_\beta^2 \mathbf{I})$	$N(0, \sigma_{\omega'}^2)$	$N(0, \sigma_{\rho'}^2)$	$N(0, \sigma_{\nu'}^2)$	$N(0, \sigma_{\gamma'}^2)$	$N(\mathbf{0}, \sigma_\psi^2 \mathbf{I})$
Hyperparameters	$\sigma_\beta^2 = 10^2$	$\sigma_{\omega'}^2 = 10^2$	$\sigma_{\rho'}^2 = 10^2$	$\sigma_{\nu'}^2 = 10^2$	$\sigma_{\gamma'}^2 = 10^2$	$\sigma_\psi^2 = 10^2$

5.1. Model comparison with full data

For full data SEMs, both DIC_1 and DIC_2 are computed to evaluate model fit, with lower values indicating better fit. DIC_1 is defined as:

$$\text{DIC}_1 = -4\mathbb{E}_\phi[\log p(\mathbf{y} | \phi)] + 2 \log p(\mathbf{y} | \bar{\phi}), \quad (21)$$

where $\bar{\phi}$ is the posterior mean of ϕ , and $p(\mathbf{y} | \phi)$ denotes the density of $\mathbf{y}$ given the model parameters ϕ . Both the expectation and the posterior mean $\bar{\phi}$ are computed using samples generated from the approximate posterior distribution estimated by the proposed VB method in Section S2 of the online supplement. For all SEMs, expressions for $\log p(\mathbf{y} | \phi)$ are presented in Section S6 of the online supplement.

DIC_2 is defined as:

$$\text{DIC}_2 = -4\mathbb{E}_\phi[\log p(\mathbf{y} | \phi)] + 2 \log p(\mathbf{y} | \hat{\phi}), \quad (22)$$

where $\hat{\phi}$ is the value of ϕ that maximises the function $p(\mathbf{y} | \phi)p(\phi)$, estimated using samples drawn from the approximate posterior obtained by the proposed VB method. The expectation is also computed using the samples from the approximate posterior.

5.2. Model comparison with missing responses

DIC_5 is used to assess model fit for SEMs with missing data and is defined as:

$$\text{DIC}_5 = -4\mathbb{E}_{\phi, \psi, \mathbf{y}_u}[\log p(\mathbf{y}, \mathbf{m} | \phi, \psi)] + 2 \log p(\mathbf{y}_o, \hat{\mathbf{y}}_u, \mathbf{m} | \hat{\phi}, \hat{\psi}), \quad (23)$$

where $p(\mathbf{y}, \mathbf{m} | \phi, \psi)$ is the likelihood of $\mathbf{y}$ and $\mathbf{m}$, conditional on model parameters ϕ and ψ . The values $\hat{\phi}$, $\hat{\psi}$, and $\hat{\mathbf{y}}_u$ are the posterior samples of ϕ , ψ and $\mathbf{y}_u$ that maximise the function $p(\mathbf{y}, \mathbf{m} | \phi, \psi)p(\phi, \psi)$, where the samples are drawn from the approximate posterior distribution estimated by the proposed HVB method in Section 4.2. The expectation is also computed using these variational posterior samples. Note that $p(\mathbf{y}, \mathbf{m} | \phi, \psi) = p(\mathbf{y} | \phi)p(\mathbf{m} | \mathbf{y}, \psi)$; see the selection model factorisation in Eq. (13) in Section 3.2.

It is worth noting that the density function $p(\mathbf{y} | \phi)$, which appears in all DIC expressions in Eqs. (21), (22), and (23), represents the density of the response vector $\mathbf{y}$ conditional on ϕ . For the SEM-Gau and YJ-SEM-Gau, this distribution is equivalent to $p(\mathbf{y} | \xi)$, as for these two models $\xi = \phi$ (see Section 2). In contrast, for the SEM-t and YJ-SEM-t, $p(\mathbf{y} | \phi)$ needs to be derived. This is because for the latter two models, $\xi = (\phi^\top, \tau^\top)^\top$, so $p(\mathbf{y} | \phi)$ represents the marginal density of $\mathbf{y}$ given ϕ , with the latent vector τ integrated out (see Section S6 of the online supplement). Further details, including the logarithm of the density $p(\mathbf{y} | \phi)$ for all four models, are provided in Section S6 of the online supplement.

6. Simulation study: Assessing the accuracy of VB methods

This section compares the accuracy of the proposed VB methods (with full data and with missing data) against the Hamiltonian Monte Carlo (HMC) (Neal, 2011) method. The HMC method is implemented using the RStan interface (Stan Development Team, 2023), specifically employing the No-U-Turn Sampler (NUTS) algorithm (Hoffman and Gelman, 2014). The posterior densities estimated using HMC are considered as ground truth to assess the accuracy of the VB methods. A relatively small simulated dataset ($n = 625$) is used for this comparison, as HMC becomes computationally expensive for SAR models with larger datasets, particularly in the presence of missing data (Wijayawardhana et al., 2025). The simulation study presented in Section 7 and the real data application in Section 8 use moderately large datasets ($n = 4,378$).

The prior distributions used for both the VB and HMC algorithms throughout this paper are described next. To map the parameters σ_e^2 , ρ , ν , and γ onto the real line, we apply the following transformations: $\omega' = \log(\sigma_e^2)$, $\rho' = \log(1 + \rho) - \log(1 - \rho)$, $\nu' = \log(\nu - 3)$, and $\gamma' = \log(\gamma) - \log(2 - \gamma)$. The prior distributions are then specified on the transformed parameter space. Table 3 summarises the priors and hyperparameters. We use diffuse (non-informative) priors for all model parameters so that posterior inference is driven primarily by the data. For example, the fixed effects β are assigned weakly informative normal priors, $\beta \sim N(\mathbf{0}, 100\mathbf{I})$, where $\mathbf{I}$ is an identity matrix of appropriate dimension (see Table 3). For the relatively large datasets analysed in this paper, the choice of these non-informative hyperparameters has little impact on the parameter estimates or on the convergence behaviour of the VB and HMC algorithms.

The comparison in this section is based on a simulated dataset generated from the YJ-SEM-Gau. The dataset contains 625 observations with five covariates, all generated from standard normal distributions. The spatial weight matrix, $\mathbf{W}$, is based on a 25×25 regular lattice, where neighbouring units are defined using the rook neighbourhood (Lloyd, 2010). The model parameters are specified as follows: the fixed effects (β) are randomly drawn from a discrete uniform distribution over -3 to 3 (excluding 0).

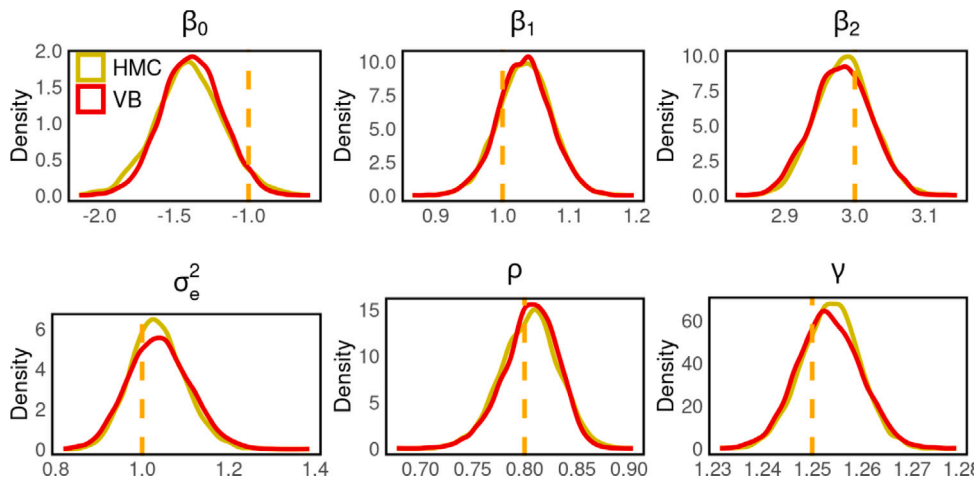


Fig. 1. Comparison of posterior densities of selected parameters in the YJ-SEM-Gau, obtained using the HMC and VB methods on the simulated dataset without missing values. The vertical line represents the true parameter value.

The error variance is set to $\sigma_e^2 = 1$ and the spatial autocorrelation parameter is $\rho = 0.8$. The Yeo–Johnson parameter is set to $\gamma = 1.25$, introducing moderate left skewness into the simulated response variable. See Figure S1 in Section S7 of the online supplement for the kernel density plot of the response variable. Section 6.1 compares the proposed VB method with HMC using this dataset without missing values, while Section 6.2 compares the proposed HVB method with HMC in the presence of missing responses by introducing missing values into this dataset.

6.1. SEMs with full data

We now assess the accuracy of the VB method for SEMs with full data. The VB algorithm, outlined in Algorithm S1 in Section S2 of the online supplement, and the HMC algorithm are used to fit the YJ-SEM-Gau to the simulated dataset. The initial values for the VB algorithm are set as follows: the variational mean vector (μ) is initialised using the estimates of β , σ_e^2 , and ρ obtained from fitting the SEM-Gau via maximum likelihood (ML) estimation method. The variational mean corresponding to the parameter γ is initialised to 1. For the variational covariance matrix parameters, the elements of $\mathbf{B}$ and the diagonal elements of $\mathbf{D}$ are all initialised to 0.01. A value of $p = 4$ factors is used. The results do not improve when the number of factors is increased. The HMC algorithm is initialised in the same way, with its initial parameter values set to the initial variational means described above. Both the VB and HMC algorithms are run for 10,000 iterations, at which point convergence is achieved. For the HMC algorithm, the first 5000 iterations are treated as burn-in and discarded. Convergence for the VB algorithm is assessed visually by inspecting the plots of variational means over iterations. For the HMC algorithms, convergence is assessed by inspecting trace plots of the model parameters (see Section S11.1 of the online supplement).

The HMC algorithm directly generates posterior samples of the model parameters. In contrast, the VB algorithm first estimates the variational parameters, λ , by running the variational optimisation. Then, 10,000 posterior draws of the model parameters, $\xi = \phi = (\beta^\top, \sigma_e^2, \rho, \gamma)^\top$, are generated from the variational distribution $q_\lambda(\xi)$; see Section S2 of the online supplement.

Fig. 1 compares the posterior densities of selected model parameters estimated using the VB and HMC algorithms. For all model parameters, the posterior densities obtained from both methods are almost indistinguishable and successfully recover the true values.

6.2. SEMs with missing responses

Now, missing values are introduced into the response variable $\mathbf{y}$ under the MNAR mechanism using the logistic model described in Eq. (11), in order to compare the performance of the HVB algorithm presented in Section 4.2 with HMC. To serve as a covariate in the logistic model, an additional variable $\mathbf{x}^*$ is generated from a log-normal distribution with log-scale mean 0 and log-scale standard deviation 1. This simulated covariate, and the response variable $\mathbf{y}$ from the YJ-SEM-Gau, are used as the predictors in the missing value model defined in Eq. (11). The coefficients are set as follows: $\psi_0 = -1.0$ (intercept), $\psi_{x^*} = 0.5$ (coefficient of $\mathbf{x}^*$), and $\psi_y = -0.1$ (coefficient of $\mathbf{y}$). This configuration yields approximately 50% missing values in $\mathbf{y}$, corresponding to $n_u = 320$.

The HVB-NoB algorithm (without block-wise update) described in Algorithm 1 and the HMC algorithm are used to estimate the YJ-SEM-Gau with missing data. The tuning parameter, N_1 of the HVB-NoB algorithm, is set to 10. The initial values for the algorithm are set as follows: the variational mean vector (μ) is initialised using the estimates of β , σ_e^2 , and ρ obtained by fitting the SEM-Gau via ML to the complete cases only, i.e., ignoring locations with missing responses. The initial value of the YJ transformation parameter γ is set to 1. All fixed-effect parameters of the missing value model (ψ) are initialised to 0.1. For the variational covariance matrix, all elements of $\mathbf{B}$ and the diagonal elements of $\mathbf{D}$ are initialised to 0.01. The missing responses ($\mathbf{y}_u$) are initialised by drawing samples

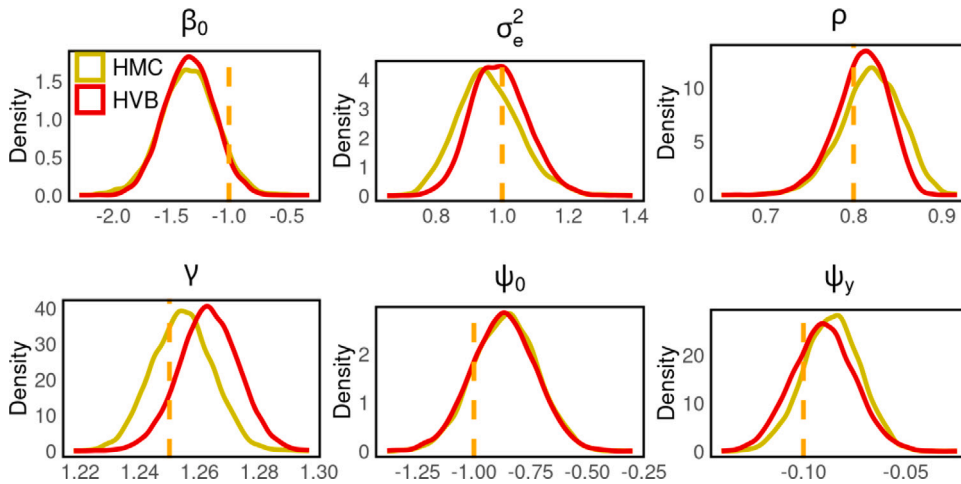


Fig. 2. Comparison of posterior densities for selected parameters of the YJ-SEM-Gau and missing data models, estimated using the HMC and HVB-NoB methods on the simulated dataset with missing values. The vertical line represents the true parameter value.

from the conditional distribution $p(\mathbf{y}_u | \xi^{(0)}, \mathbf{y}_o)$, where $\xi^{(0)}$ denotes the vector of initial parameter values, set to the corresponding initial variational means described above. We use $p = 4$ factors in the HVB-NoB algorithm, as increasing the number of factors did not lead to improved performance. The HMC algorithm is initialised similarly: initial parameter values are set in the same way as the variational means, and missing values are initialised using draws from $p(\mathbf{y}_u | \xi^{(0)}, \mathbf{y}_o)$. Both the HVB-NoB and HMC algorithms are run for 10,000 iterations. Convergence diagnostics plots are provided in Section S11.1 of the online supplement.

Now, we briefly explain how posterior samples of the model parameters, ξ , ψ , and the missing values $\mathbf{y}_u$, are generated in the HVB-NoB method. Once the algorithm converges, the set of variational parameters $\lambda = (\mu^\top, \text{vech}(\mathbf{B})^\top, \mathbf{d}^\top)^\top$ is obtained. Given these variational parameters, 10,000 draws from the variational distribution

$$q_\lambda^0(\xi, \psi) = N((\xi^\top, \psi^\top)^\top; \mu, \mathbf{B}\mathbf{B}^\top + \mathbf{D}^2)$$

are generated, forming the posterior samples of ξ and ψ . Conditional on these parameter samples, posterior samples of $\mathbf{y}_u^{(i)}$ are drawn from

$$p(\mathbf{y}_u | \mathbf{y}_o, \mathbf{m}, \xi^{(i)}, \psi^{(i)}),$$

using the MCMC steps described in Algorithm 2, where $\xi^{(i)}$ and $\psi^{(i)}$ denote the i th posterior draws of the model parameters, for $i = 1, \dots, 10,000$.

Fig. 2 compares the posterior densities of selected model parameters from the YJ-SEM-Gau and the missing data model, estimated using HVB-NoB and HMC on the simulated dataset with missing values. For most parameters, the posterior densities produced by the two methods are nearly identical. The posterior density of γ estimated using HVB-NoB is slightly different compared to HMC, which is expected since HVB is an approximate inference method, whereas HMC yields asymptotically exact posterior samples. Importantly, this minor loss in accuracy is offset by the substantial computational efficiency gained with HVB-NoB, as discussed in Section 6.3.

6.3. Summary and computation time comparison

Overall, the simulation results demonstrate that the proposed VB methods, both with full data and with missing data, provide accurate approximations to the posterior distributions obtained via HMC. For all key model parameters, the VB-based posterior densities closely match those from HMC, with only a slight overestimation observed for the transformation parameter γ in the missing data setting.

Notably, the computation times per iteration of the VB and HVB-NoB algorithms are significantly lower than those of the HMC algorithm. The computation time per iteration for the VB algorithm is approximately 0.0361 s, while for the HMC algorithm, it is 0.4680 s when estimating the YJ-SEM-Gau without missing data. The computation time per iteration for the HVB-NoB algorithm is approximately 0.0542 s, while for the HMC algorithm, it is 41.1658 s for estimating the YJ-SEM-Gau with missing data. These results demonstrate that the proposed VB and HVB-NoB methods provide accurate parameter estimates while offering substantial computational savings relative to HMC.

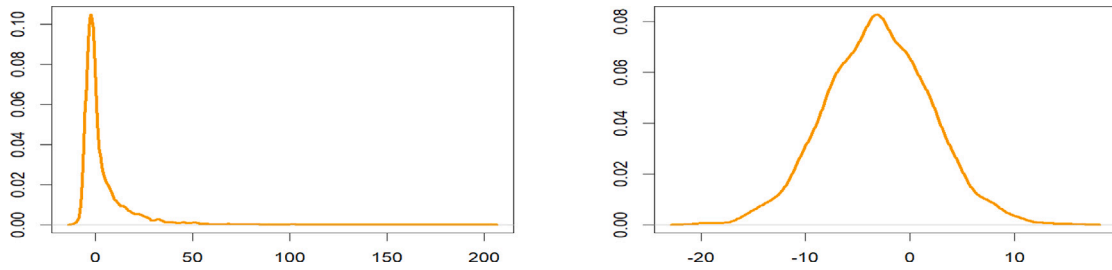


Fig. 3. Kernel density plots of the simulated data: The left panel displays the kernel density of the response variable from the YJ-SEM-t with $\nu = 4$ and $\gamma = 0.5$, displaying right skewness and heavy tails (dataset 1). The right panel displays the kernel density of the response variable from the YJ-SEM-t with $\nu = 30$ and $\gamma = 1$, which appears nearly symmetric with light tails (dataset 2).

7. Simulation study: Assessing the accuracy and robustness of proposed SEMs

This section evaluates the robustness and accuracy of the proposed SEMs and the VB methods using moderately large simulated datasets. The objective of this simulation study is to assess how well the proposed SEMs capture key characteristics of non-Gaussian spatial data, such as skewness and heavy-tailed distributions, both with full data and in the presence of missing values.

We generate datasets using a spatial weight matrix from the `spData` R package (Bivand et al., 2023), based on 25,357 houses sold between 1993 and 1998 in Lucas County, Ohio, USA (see Section 8 for further details). We simulate two datasets from the YJ-SEM-t using the spatial weight matrix $\mathbf{W}_{1998}$, corresponding to 4378 houses sold in 1998:

1. Dataset 1: $\gamma = 0.5$ (right skew) and $\nu = 4$ (heavy tails).
2. Dataset 2: $\gamma = 1$ (no skew) and $\nu = 30$ (light tails).

The remaining parameters for both datasets are set as follows: six fixed effect parameters (β) are randomly drawn from a discrete uniform distribution between -3 and 3 (excluding 0), $\sigma_e^2 = 0.5$, $\rho = 0.8$, and all covariates are generated from a standard normal distribution, $N(0, 1)$. Fig. 3 displays kernel density plots of the response variables for both datasets.

Section S8 of the online supplement fits the SEM-Gau, SEM-t, YJ-SEM-Gau, and YJ-SEM-t to simulated datasets 1 and 2 with complete data and evaluates how effectively these models, along with their VB estimation methods, capture the underlying characteristics of the datasets. In Section 7.1, missing values are introduced, and the performance of the SEMs along with the HVB estimation method (described in Section 4.2) is assessed under missing data.

7.1. SEMs with missing responses

This section evaluates the robustness and accuracy of the proposed SEMs and the HVB method in the presence of missing responses, using simulated datasets 1 and 2. First, missing values are introduced in the response variable $\mathbf{y}$ of each simulated dataset under the MNAR mechanism using the logistic regression model in Eq. (11). To construct this model, an additional covariate $\mathbf{x}^*$ is generated from a log-normal distribution with a log-scale mean of 0 and a log-scale standard deviation of 1 , and the model is regressed on $\mathbf{x}^*$ and the response variable $\mathbf{y}$. For the dataset 1, the logistic regression parameters are set to $\psi_0 = 1$ (intercept), $\psi_{\mathbf{x}^*} = -1$ (the coefficient for $\mathbf{x}^*$), and $\psi_{\mathbf{y}} = -0.1$ (the coefficient for $\mathbf{y}$). This configuration results in approximately 40% missing responses, leading to $n_u = 1,729$. For the dataset 2, the intercept is set to $\psi_0 = 0.5$ (with the other coefficients unchanged), leading to approximately 40% missing responses ($n_u = 1,774$).

We fit SEM-Gau, SEM-t, YJ-SEM-Gau, and YJ-SEM-t to these simulated datasets with missing values. As n and n_u are large, the HVB-AllB algorithm, which employs the MCMC steps detailed in Algorithm S2 in Section S5.4 of the online supplement, is utilised. The tuning parameters for the HVB-AllB algorithm are set as follows: the number of MCMC iterations (N_1) is set to 10 , and the block size (k^*) is defined as 10% of n_u , resulting in 11 blocks. The initial values for the variational mean vector $\boldsymbol{\mu}$ corresponding to β , σ_e^2 , ρ , γ , as well as the initial values for the missing responses $\mathbf{y}_u$, are specified using the procedure described in Section 6.2. For models with Student's t errors, the degrees of freedom parameter ν and the set of latent variables $\boldsymbol{\tau} = (\tau_1, \dots, \tau_n)^T$ must also be initialised. The variational mean corresponding to ν is initialised at 4 . The variational means corresponding to the latent variables are initialised by sampling from the prior distribution $p(\tau_i | \nu^{(0)}) \sim \text{IG}(\frac{\nu^{(0)}}{2}, \frac{\nu^{(0)}}{2})$ for $i = 1, \dots, n$, where $\nu^{(0)} = 4$ denotes the initial value of ν . For the variational covariance matrix, all elements of $\mathbf{B}$ and the diagonal elements of $\mathbf{D}$ are initialised to 0.01 .

We run the HVB-AllB algorithms for 10,000 iterations for the SEM-Gau and YJ-SEM-Gau across both simulated datasets. For the SEM-t and YJ-SEM-t, convergence generally requires more iterations due to the additional complexity introduced by the latent variables $\boldsymbol{\tau}$ and the parameter ν . However, the number of required iterations varies across datasets and models. For simulated dataset 1, SEM-t is run for 20,000 iterations, whereas YJ-SEM-t achieves convergence within 10,000 iterations. For simulated dataset 2, SEM-t requires 30,000 iterations, while YJ-SEM-t is run for 20,000 iterations. Convergence diagnostic plots are provided in Section S11.2 of the online supplement.

Table 4

Posterior means and 95% credible intervals for selected parameters of different SEMs based on the simulated dataset 1 with missing data, along with the computation time (CT) in seconds for one VB iteration. The table also presents the DIC₅ values for each model. Parameters labelled as 'NA' indicate that they are not applicable to the corresponding model.

	SEM-Gau	SEM-t	YJ-SEM-Gau	YJ-SEM-t
$\beta_0 = -1$	1.9268 (1.6342, 2.2270)	1.2929 (1.1456, 1.4382)	-0.9862 (-1.0256, -0.9462)	-0.9680 (-1.0238, -0.9123)
$\beta_1 = 1$	1.8848 (1.6344, 2.1381)	1.3230 (1.1675, 1.4797)	0.9938 (0.9554, 1.0332)	0.9835 (0.9338, 1.0324)
$\sigma_e^2 = 0.5$	77.4912 (74.2316, 80.7856)	24.4549 (23.2015, 25.7503)	0.9994 (0.9331, 1.0701)	0.7028 (0.6615, 0.7451)
$\rho = 0.8$	0.2839 (0.2527, 0.3144)	0.1948 (0.1722, 0.2177)	0.8007 (0.7858, 0.8149)	0.8044 (0.7922, 0.8159)
$\nu = 4$	NA	3.3400 (3.2319, 3.4772)	NA	9.5082 (9.0327, 10.0109)
$\gamma = 0.5$	NA	NA	0.4965 (0.4822, 0.5113)	0.4990 (0.4882, 0.5099)
$\psi_0 = 1$	0.8251 (0.7048, 0.9441)	0.9171 (0.8126, 1.0203)	0.9337 (0.8193, 1.0442)	0.9515 (0.8525, 1.0522)
$\psi_y = -0.1$	-0.0793 (-0.0907, -0.0679)	-0.0997 (-0.1178, -0.0812)	-0.0794 (-0.0924, -0.0663)	-0.0790 (-0.0891, -0.0686)
DIC ₅	36009.5	36302.7	18278.55	17855.59
CT	0.5894	1.2250	0.6000	1.339

7.1.1. Results for the simulated dataset 1

This section presents the results for dataset 1 with missing values. Table 4 presents the posterior means and 95% credible intervals for some of the model parameters of the SEM-Gau, YJ-SEM-Gau, SEM-t, and YJ-SEM-t obtained using the HVB-AllB method applied to the simulated dataset 1 with missing values, along with the computation cost for one iteration. The table also includes the DIC₅ values for each model, calculated using the formula in Eq. (23). Corresponding posterior density plots of the parameters are given in Figure S4 in Section S9.1 of the online supplement.

We start by comparing the estimated values of the fixed effects (β_0 and β_1 for the SEMs and ψ_0 and ψ_1 for the missing data model), the variance parameter σ_e^2 , and the spatial autocorrelation parameter ρ , as these parameters are common to all four SEMs. The posterior means for β_0 , β_1 , ψ_0 , ψ_y , and ρ are nearly identical for both YJ-SEM-Gau and YJ-SEM-t, closely matching the true values. In contrast, the estimates from SEM-Gau and SEM-t show significant deviations from the true values. The estimated posterior means of σ_e^2 from SEM-Gau, SEM-t, and YJ-SEM-Gau differ considerably from the true value, while the posterior mean from YJ-SEM-t is much closer to the true value. The models incorporating the YJ transformation (YJ-SEM-Gau and YJ-SEM-t) successfully recover the true value of the parameter γ . The estimates of ν obtained from the SEM-t and YJ-SEM-t are slightly inaccurate. Additionally, the posterior means of the estimated missing values from YJ-SEM-t and YJ-SEM-Gau align more closely with the true missing values compared to those from SEM-Gau and SEM-t; see Figure S6 in Section S9.2 of the online supplement.

For the simulated dataset 1 with missing values in the response variable, the YJ-SEM-t provides the best fit, as indicated by the lowest DIC₅ value, among all four models. This result is expected, as the dataset is generated from the YJ-SEM-t, which incorporates both heavy tails and skewness. The second-best performer is YJ-SEM-Gau, which has the next lowest DIC₅ value. Furthermore, similar to the case without missing values discussed in Section S8 of the online supplement, the relatively small difference in the DIC₅ values between YJ-SEM-t and YJ-SEM-Gau suggests that YJ-SEM-Gau provides a comparably good fit for dataset 1.

The left four panels of Fig. 4 compare the kernel density of the true missing values (y_u) with the kernel densities of the posterior means of the missing values estimated under different SEMs for the simulated dataset 1. The densities from both YJ-SEM-Gau and YJ-SEM-t are nearly identical and closely align with the density of the true missing values. The right panel of Fig. 4 presents the posterior density of the maximum missing value, $\max(y_u)$, estimated under each SEM. The true value of the maximum missing value lies closer to the posterior densities produced by YJ-SEM-Gau and YJ-SEM-t than to those produced by SEM-Gau and SEM-t. This behaviour is expected, as YJ-SEM-Gau and YJ-SEM-t provide parameter estimates that are closer to the true values and yield more comparable DIC₅ values (see Table 4).

7.1.2. Results for the simulated dataset 2

This section presents the results for dataset 2 with missing values. Table 5 reports the posterior means and 95% credible intervals for some of the model parameters of the SEM-Gau, SEM-t, YJ-SEM-Gau, and YJ-SEM-t, obtained using the HVB-AllB method on the simulated dataset 2 with missing values, along with the computation cost for one VB iteration. The table also presents the corresponding DIC₅ values for each model. Posterior density plots for these parameters are provided in Figure S4 in Section S9.1 of the online supplement. The estimated posterior means for β_0 , β_1 , σ_e^2 , ρ , ψ_0 , and ψ_y across all four models closely match their true values. Similarly, the estimated posterior mean of γ for the YJ-SEM-Gau and YJ-SEM-t is also close to the true value. The parameter ν is accurately estimated for both the SEM-t and YJ-SEM-t. In addition, all four models produce posterior means for the estimated missing values that closely match the true values; see Figure S6 in Section S9.2 of the online supplement.

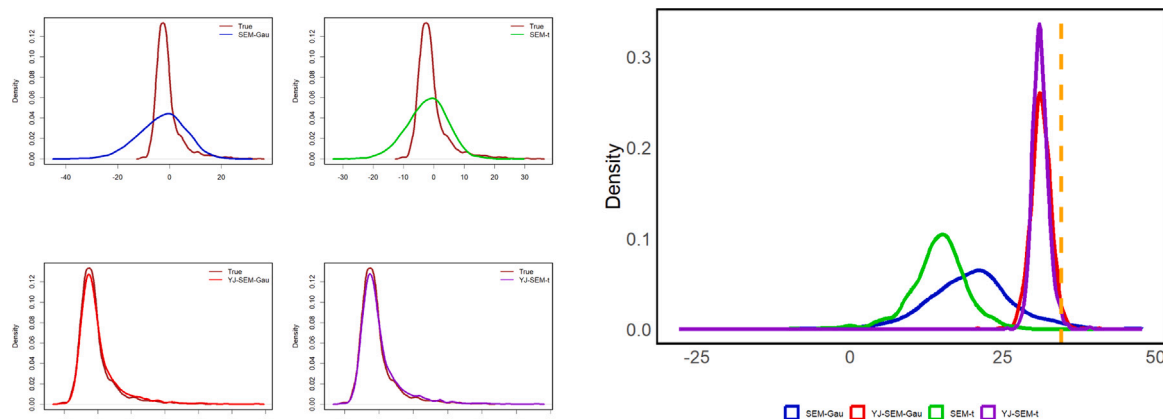


Fig. 4. Left panel: The kernel density of the true missing values (y_u) and the kernel densities of posterior means of the missing values obtained using the HVB-AllB method for different SEMs for the simulated dataset 1 with missing values. Right panel: The posterior density of the maximum missing value, $\max(y_u)$, obtained by different SEMs for the simulated dataset 1 with missing values. The true value of the maximum missing value is indicated by the vertical line.

Table 5

Posterior means and 95% credible intervals for selected parameters of different SEMs based on the simulated dataset 2 with missing data, along with the computation time (CT) in seconds for one VB iteration. The table also presents the DIC_5 values for each model. Parameters labelled as ‘NA’ indicate that they are not applicable to the corresponding model.

	SEM-Gau	SEM-t	YJ-SEM-Gau	YJ-SEM-t
$\beta_0 = -3$	-3.0131 (-3.0495, -2.9764)	-3.0150 (-3.0497, -2.9792)	-2.9511 (-2.9820, -2.9199)	-2.9467 (-2.9767, -2.9161)
$\beta_1 = 2$	1.9942 (1.9582, 2.0309)	1.9884 (1.9605, 2.0165)	1.9803 (1.9621, 1.9979)	1.9834 (1.9568, 2.0096)
$\sigma_e^2 = 0.5$	0.5528 (0.5222, 0.5849)	0.5143 (0.4892, 0.5406)	0.5450 (0.5155, 0.5759)	0.5096 (0.4826, 0.5377)
$\rho = 0.8$	0.8002 (0.7857, 0.8141)	0.7990 (0.7874, 0.8101)	0.8003 (0.7892, 0.8113)	0.7971 (0.7841, 0.8095)
$\nu = 30$	NA	30.2880 (29.1096, 31.4718)	NA	29.7195 (27.9820, 31.6032)
$\gamma = 1$	NA	NA	1.0032 (0.9933, 1.0129)	1.0017 (0.9961, 1.0074)
$\psi_0 = 0.5$	0.4949 (0.3847, 0.6044)	0.5036 (0.4219, 0.5841)	0.5477 (0.4319, 0.6619)	0.5467 (0.4742, 0.6204)
$\psi_\gamma = -0.1$	-0.0956 (-0.1210, -0.0706)	-0.1090 (-0.1320, -0.0864)	-0.0965 (-0.1106, -0.0827)	-0.0972 (-0.1137, -0.0809)
DIC_5	14713.89	14221.78	14695.32	14164.86
CT	0.5755	1.2166	0.6048	1.3486

Based on the DIC_5 values, the YJ-SEM-t provides the best fit for simulated dataset 2 with missing values, while the SEM-t has the second-lowest DIC_5 , indicating it is the second-best performing model.

The kernel densities of the posterior means of the missing values (y_u) estimated under all four SEMs closely match the kernel density of the true missing values for simulated dataset 2; see left panel of Figure S8 in Section S9.2 of the online supplement. Similarly, for each SEM, the posterior density of the maximum missing value, $\max(y_u)$, contains the true value, and the densities across the different SEMs are nearly identical; see the right panel of Figure S8 in Section S9.2 of the online supplement. These results are consistent with expectations, as all four models produce similar parameter estimates and DIC_5 values, with estimates generally close to the true parameter values (see Table 5).

7.2. Summary and computation time comparison

This simulation study evaluates how well the proposed SEMs capture key characteristics of non-Gaussian spatial data, both with fully observed data and in the presence of missing values, using two simulated datasets: dataset 1 exhibits skewness and heavy tails, while dataset 2 is closer to Gaussian. This section summarises the findings of the simulation study with missing data, while Section S8.3 of the online supplement summarises the corresponding results for the full data.

As shown in Tables 4 and 5, the HVB-AllB algorithm applied to the SEM-t and YJ-SEM-t, both of which assume Student’s t -distributed errors, requires more computation time per iteration compared to SEM-Gau and YJ-SEM-Gau. This is due to the added

complexity of VB optimisation: these models introduce an additional latent variable vector of length n and include the degrees of freedom parameter ν , increasing the dimensionality of the optimisation. In addition, SEM-t and YJ-SEM-t typically require more VB iterations to converge. Convergence diagnostics plots in Section S11.2 of the online supplement illustrate this behaviour. Consequently, HVB-AllB for SEM-t and YJ-SEM-t is generally more time-consuming than for SEM-Gau and YJ-SEM-Gau, both because each iteration is slower and more iterations are needed to achieve convergence.

In terms of performance, YJ-SEM-Gau and YJ-SEM-t are particularly effective for modelling skewed and heavy-tailed data. For simulated dataset 1, YJ-SEM-t attains the lowest DIC₅ value, indicating the best fit, while YJ-SEM-Gau achieves the second-lowest value, which is close to that of YJ-SEM-t. For simulated dataset 2, which is closer to Gaussian, all four models recover the true parameter values accurately and yield nearly identical DIC₅ values, with YJ-SEM-t again achieving the lowest. Although estimating YJ-SEM-t incurs substantially higher computational cost compared to SEM-Gau (the conventional SEM with Gaussian errors), YJ-SEM-Gau maintains a computation time comparable to SEM-Gau while providing considerable improvements when modelling non-Gaussian data. Similar patterns are observed in the analysis of the fully observed datasets, as presented in Section S8 of the online supplement.

8. Real data application

We apply the proposed SEMs and VB methods to a dataset of single-family homes sold in Lucas County, Ohio, USA. This dataset is available in the R package *spData* (Bivand et al., 2023). The data set includes various house characteristics: house age, the lot size in square feet, the number of rooms, the total living area in square feet, the number of bedrooms, and binary indicators for each year from 1993 to 1998 to represent the year of house sale. For our analysis, we focus on the house price data from 1998, which contains 4378 observations. We call this dataset Lucas-1998-HP. House prices are typically positive and often exhibit skewness. To transform house prices into the entire real line and reduce skewness, researchers commonly apply logarithmic transformations before modelling (LeSage and Pace, 2004; Wijayawardhana et al., 2024). Accordingly, we apply the natural logarithmic transformation to the house prices in the Lucas-1998-HP dataset. However, even after the transformation, the resulting log-prices remain left-skewed; see Figure S9 in Section S10 of the online supplement.

First, the proposed VB algorithm (Algorithm S1 in Section S2 of the online supplement) is used to estimate all four SEMs using the full Lucas-1998-HP dataset. Next, missing values are introduced into the response variable, and the SEMs are re-estimated using the proposed HVB-AllB algorithm (Algorithm 1, together with the MCMC steps in Algorithm S2 in Section S5.4 of the online supplement). This design enables a direct comparison between estimates obtained from the full dataset and those obtained in the presence of missing values, with the full-data estimates treated as the ground truth. The set of predictor variables includes various powers of house age (age, age², and age³), the natural logarithm of the lot size in square feet (ln(lotsize)), the number of rooms (rooms), the natural logarithm of the total living area in square feet (ln(LTA)), and the number of bedrooms (beds). The natural logarithm of house price in hundreds of thousands of US dollars is selected as the response variable. Results for the complete dataset (without missing values) are presented in Section 8.1, while Section 8.2 provides the findings with missing data.

8.1. Full data SEMs

This section presents the results of fitting all four SEMs to the full Lucas-1998-HP dataset. The initial values for the VB algorithms are set following the same approach used in the simulation studies in Sections 6 and 7, and we use $p = 4$ factors. Consistent with the simulation results with full data (see Section S8 of the online supplement), the models with Student's t errors require more iterations to achieve convergence. Accordingly, the VB algorithms are run for 20,000 iterations for SEM-Gau and YJ-SEM-Gau, and 75,000 iterations for SEM-t and YJ-SEM-t. Convergence is assessed visually using the trajectories of the variational means, with the diagnostic plots provided in Figure S20 in Section S11.3 of the online supplement.

The estimated posterior means and 95% credible intervals for selected model parameters across different SEMs, along with their corresponding DIC values, are summarised in Table 6. The posterior density plots for these parameters are provided in Figure S10 in Section S10 of the online supplement. According to the DIC₁ and DIC₂ values, the YJ-SEM-t is the most appropriate SEM for the Lucas-1998-HP dataset, followed by the YJ-SEM-Gau as the second most suitable.

We now compare estimates of key parameters across different SEMs. A crucial aspect of SEM estimation is identifying the underlying spatial correlation, represented by the parameter ρ . For the SEM-Gau, YJ-SEM-Gau, and YJ-SEM-t, the estimated values of ρ are relatively close, with posterior mean values of 0.6291, 0.6019, and 0.5990, respectively. This suggests a moderate spatial correlation in the Lucas-1998-HP dataset. However, for the SEM-t, the estimated ρ is noticeably lower, with a posterior mean value of 0.5475, compared to the other three models.

Similar to linear regression models, the fixed effects β in SEMs represent the influence of covariates on the response variable, which in this case is the logarithm of the house price. An interesting disparity arises in the estimated posterior mean of the fixed effect for the covariate 'age' (β_{age}). For the SEM-Gau, β_{age} is positive, while for the other three models, β_{age} is negative. This discrepancy reflects a key limitation of SEM-Gau: it is misspecified for this dataset because it cannot adequately capture important features of the response variable, such as skewness and heavy tails. As a result, SEM-Gau produces misleading estimates for the effect of house age, suggesting an increase in price rather than the decrease indicated by the more flexible models. Similarly, the estimated posterior means of the fixed effect for the number of rooms (β_{rooms}) are 0.0068, 0.0188, 0.0311, and 0.0238 for SEM-Gau, SEM-t, YJ-SEM-Gau, and YJ-SEM-t, respectively, showing that the effect of the number of rooms on house prices varies across models. Notably,

Table 6

Posterior means and 95% credible intervals for selected model parameters of different SEMs for the Lucas-1998-HP dataset (without missing values). The table also includes DIC values for each model. Parameters labelled as ‘NA’ indicate that they are not applicable to the corresponding model.

	SEM-Gau	SEM-t	YJ-SEM-Gau	YJ-SEM-t
<i>intercept</i>	−0.4336 (−0.4469, −0.4207)	−0.3865 (−0.3979, −0.3752)	−0.3020 (−0.3278, −0.2765)	−0.2849 (−0.3087, −0.2612)
β_{age}	0.1650 (0.0557, 0.2742)	−0.4197 (−0.5236, −0.3162)	−0.1643 (−0.2926, −0.0321)	−0.3201 (−0.3984, −0.2405)
β_{rooms}	0.0068 (−0.0176, 0.0313)	0.0188 (−0.0018, 0.0395)	0.0311 (−0.0281, 0.0893)	0.0238 (−0.0056, 0.0524)
$\beta_{\log(\text{lotsize})}$	0.1614 (0.1480, 0.1752)	0.1193 (0.1033, 0.1350)	0.1441 (0.1068, 0.1815)	0.1306 (0.1192, 0.1420)
σ_e^2	0.1577 (0.1516, 0.1641)	0.0670 (0.0633, 0.0708)	0.1031 (0.0967, 0.1099)	0.0695 (0.0647, 0.0744)
ρ	0.6291 (0.5605, 0.6902)	0.5475 (0.4900, 0.6015)	0.6019 (0.5236, 0.6712)	0.5990 (0.5261, 0.6651)
ν	NA	3.0249 (3.0000, 3.1324)	NA	7.9482 (7.5327, 8.3940)
γ	NA	NA	1.5264 (1.4783, 1.5728)	1.5529 (1.5145, 1.5904)
DIC ₁	4470.316	4878.664	4329.431	3720.687
DIC ₂	4469.655	5031.204	4332.07	3766.714

YJ-SEM-Gau shows the strongest positive influence, while SEM-Gau exhibits the weakest, further reflecting the consequences of misspecification.

Additionally, the SEM-t, YJ-SEM-Gau, and YJ-SEM-t provide further insights into the characteristics of the dataset. The posterior means of $\nu = 3.0249$ for SEM-t and 7.9482 for YJ-SEM-t suggest that the log-transformed house price distribution exhibits heavy tails. Furthermore, the posterior means of γ , 1.5264 for YJ-SEM-Gau and 1.5529 for YJ-SEM-t, indicate left skewness in the response variable, effectively capturing its asymmetry. These distributional features are visually evident in the kernel density plot of the response variable (see Figure S9 in Section S10 of the online supplement).

8.2. SEMs with missing responses

This section analyses the Lucas-1998-HP dataset with missing values in the response variable (logarithm of the house price). Starting from the complete dataset, missing responses are generated using the logistic regression model in Eq. (11), with the age of the house and the response variable y (from the SEMs) serving as covariates. The missing data model parameters are set to $\psi_0 = 0.1$, $\psi_{x^*} = 0.1$, and $\psi_y = -0.1$. This configuration results in approximately 50% of the responses being missing, yielding $n_u = 2,348$.

All four SEMs are fitted to the Lucas-1998-HP dataset with missing values using the HVB method. Since both n and n_u are moderately large, the HVB-AllB algorithm is employed. The tuning parameters for the algorithm are set as follows: the number of MCMC iterations (N_i) is set to 10, and the block size (k^*) is chosen as 10% of n_u , resulting in 11 blocks. The initial values for the HVB-AllB algorithm are set in the same manner as in the simulation study presented in Section 7.1. The number of factors is set to 4 (i.e., $p = 4$). Similar to the simulation study with missing data described in Section 7.1, models with Student's t errors require a relatively high number of iterations to converge. Accordingly, the HVB-AllB algorithm is run for 10,000 iterations for both the SEM-Gau and YJ-SEM-Gau, 40,000 iterations for the SEM-t, and 20,000 iterations for the YJ-SEM-t. Convergence plots can be found in Figure S21 in Section S11.3 of the online supplement.

The estimated posterior means and 95% credible intervals for selected model parameters across different models, along with their corresponding DIC₅ values, are summarised in Table 7. The posterior density plots for these parameters are provided in Figure S11 in Section S10 of the online supplement. According to the DIC₅ values, the YJ-SEM-t is the most appropriate SEM for the Lucas-1998-HP dataset with missing data, followed by the YJ-SEM-Gau as the second-best model.

The estimated parameters from different SEMs are now briefly discussed. For the spatial autocorrelation parameter, ρ , the posterior mean estimates across all SEMs are slightly lower than those obtained using the full dataset without missing values, except under the SEM-t (see Table 6). However, these estimates still indicate a moderate spatial correlation in the Lucas-1998-HP dataset, consistent with the conclusions drawn from SEMs fitted without missing values in Section 8.1. The posterior mean estimates for the fixed effects (β) in the SEMs lead to the same conclusions as those drawn from the Lucas-1998-HP dataset without missing values. For example, the posterior mean estimate of β_{age} in the SEM-Gau suggests a positive effect of age on house prices, whereas the estimates from the other three models indicate a negative effect. Moreover, the estimated posterior means of intercept in the missing data model, ψ_0 , are close to the true value of 0.1 for all models except SEM-t. The posterior mean estimates of ν for the SEM-t and YJ-SEM-t indicate moderate to strong heavy-tailed behaviour. Similarly, the estimated posterior mean values of γ for the YJ-SEM-Gau and YJ-SEM-t suggest left skewness in the response. These conclusions are consistent with the findings from the analysis without missing data discussed in Section 8.1.

Table 7

Posterior means and 95% credible intervals for selected parameters of different SEMs for the Lucas-1998-HP dataset with missing data. The table also includes DIC₅ values for each model. Parameters labelled as 'NA' indicate that they are not applicable to the corresponding model.

	SEM-Gau	SEM-t	YJ-SEM-Gau	YJ-SEM-t
<i>intercept</i>	−0.4189 (−0.4413, −0.3960)	−0.3721 (−0.3886, −0.3562)	−0.2856 (−0.3008, −0.2703)	−0.2752 (−0.2956, −0.2544)
β_{age}	0.1935 (0.0335, 0.3510)	−0.4640 (−0.5990, −0.3324)	−0.1473 (−0.2598, −0.0345)	−0.3768 (−0.4870, −0.2687)
β_{rooms}	0.0010 (−0.0409, 0.0432)	0.0179 (−0.0132, 0.0491)	0.0341 (0.0021, 0.0655)	0.0438 (0.0053, 0.0816)
$\beta_{\log(\text{lotsize})}$	0.1360 (0.1097, 0.1620)	0.1124 (0.0952, 0.1293)	0.1173 (0.1020, 0.1323)	0.1219 (0.0786, 0.1651)
σ_e^2	0.1463 (0.1378, 0.1553)	0.0696 (0.0649, 0.0746)	0.0894 (0.0847, 0.0943)	0.0657 (0.0617, 0.0699)
ρ	0.5872 (0.4810, 0.6818)	0.5686 (0.4553, 0.6695)	0.5494 (0.4337, 0.6502)	0.5377 (0.4325, 0.6322)
ν	NA	4.1165 (3.9476, 4.3046)	NA	10.6394 (10.2060, 11.1003)
γ	NA	NA	1.5814 (1.5297, 1.6299)	1.5998 (1.5552, 1.6415)
$\psi_0 = 0.1$	0.1128 (0.0193, 0.2064)	0.2075 (0.1361, 0.2806)	0.1110 (0.0349, 0.1865)	0.1317 (0.0595, 0.2033)
DIC ₅	8074.382	7734.356	7309.394	6657.172

9. Conclusion

Our article introduces three novel simultaneous autoregressive (SAR) models for non-Gaussian spatial data. Specifically, we extend the conventional spatial error model (SEM-Gau), a widely used SAR model that assumes Gaussian errors, to accommodate skewed and heavy-tailed features of real datasets. The novel SEMs are SEM-t, YJ-SEM-Gau, and YJ-SEM-t. The SEM-t assumes that the error terms follow a Student's t distribution, allowing for heavier tails than the Gaussian case. The YJ-SEM-Gau retains Gaussian errors but applies the Yeo–Johnson (YJ) transformation to handle asymmetric response variables. The YJ-SEM-t combines these approaches by assuming Student's t errors while incorporating the YJ transformation. These extensions improve the robustness of SAR models for analysing non-Gaussian spatial datasets. Additionally, we develop efficient variational Bayes (VB) methods to estimate these models with and without missing data. In cases with missing data, we assume that the responses are missing under the missing not at random (MNAR) mechanism. Standard VB methods are inadequate for accurately estimating these SEMs in the presence of missing data; therefore, we employ hybrid VB (HVB) algorithms to address this challenge. The posterior distributions obtained from the proposed VB and HVB methods are compared with those of the exact Hamiltonian Monte Carlo (HMC) method, demonstrating both accuracy and substantial computational gains. Although the proposed model extensions are illustrated using the SEM, both the modelling framework and the associated VB/HVB inference methods are general and can be readily adapted to other SAR-type models.

The empirical results show that: (1) The VB and HVB methods yield posterior density estimates for most model parameters that are similar to those obtained using the HMC method when estimating SEMs with and without missing values; (2) For models with Student's t errors (SEM-t and YJ-SEM-t), the posterior approximation of the degrees of freedom parameter (ν) from the VB and HVB methods exhibits slight deviations compared to the true value; (3) For both VB and HVB algorithms, models with Student's t errors require more computation time per iteration than models with Gaussian errors (SEM-Gau and YJ-SEM-Gau); (4) In general, models with Gaussian errors tend to converge in fewer iterations compared to those with Student's t errors; (5) YJ-SEM-t is able to model complex response variables with skewness and heavy tails, and it provides the best fit to the Lucas-1998-HP dataset, both with and without missing data.

Future work could explore more flexible transformation families, such as Tukey's g-and-h transformation (Tukey, 1977; Martinez and Iglewicz, 1984), which jointly accommodate skewness and heavy tails. In addition, while the proposed non-Gaussian SEMs are developed for univariate spatial responses, extending the framework to multivariate and spatio-temporal settings would enable joint modelling of multiple non-Gaussian variables and dynamic processes. These extensions would substantially broaden the applicability of the models to a wider range of real-world spatial and spatio-temporal datasets.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.spasta.2026.100966>.

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